

EECS 841

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Convolution Results for Each Channel

1. Convolution of $I[0, :, :]$ with $F[0, :, :]$

Input Matrix $I[0, :, :]$:

$$\begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 1 & -2 & -3 \\ -1 & 3 & 4 & 0 \end{bmatrix}$$

Filter $F[0, :, :]$:

$$\begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

Output size: 2×3

Convolution Steps:

$$\text{Output}[0, 0] = 4(1) + 2(-2) + 6(0) + 1(1) = 4 - 4 + 0 + 1 = 1$$

$$\text{Output}[0, 1] = 2(1) + 1(-2) + 1(0) + (-2)(1) = 2 - 2 + 0 - 2 = -2$$

$$\text{Output}[0, 2] = 1(1) + 3(-2) + (-2)(0) + (-3)(1) = 1 - 6 + 0 - 3 = -8$$

$$\text{Output}[1, 0] = 6(1) + 1(-2) + (-1)(0) + 3(1) = 6 - 2 + 0 + 3 = 7$$

$$\text{Output}[1, 1] = 1(1) + (-2)(-2) + 3(0) + 4(1) = 1 + 4 + 0 + 4 = 9$$

$$\text{Output}[1, 2] = -2(1) + (-3)(-2) + 4(0) + 0(1) = -2 + 6 + 0 + 0 = 4$$

Final Output Matrix:

$$\begin{bmatrix} 1 & -2 & -8 \\ 7 & 9 & 4 \end{bmatrix}$$

2. Convolution of $I[1, :, :]$ with $F[1, :, :]$

Input Matrix $I[1, :, :]$:

$$\begin{bmatrix} -2 & 1 & 4 & 5 \\ 0 & 6 & 3 & 6 \\ 4 & -1 & 1 & 1 \end{bmatrix}$$

Filter $F[1, :, :]$:

$$\begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$$

Output size: 2×3

Convolution Steps:

$$\text{Output}[0, 0] = -2(1) + 1(0) + 0(1) + 6(-1) = -2 + 0 + 0 - 6 = -8$$

$$\text{Output}[0, 1] = 1(1) + 4(0) + 6(1) + 3(-1) = 1 + 0 + 6 - 3 = 4$$

$$\text{Output}[0, 2] = 4(1) + 5(0) + 3(1) + 6(-1) = 4 + 0 + 3 - 6 = 1$$

$$\text{Output}[1, 0] = 0(1) + 6(0) + 4(1) + (-1)(-1) = 0 + 0 + 4 + 1 = 5$$

$$\text{Output}[1, 1] = 6(1) + 3(0) + (-1)(1) + 1(-1) = 6 + 0 - 1 - 1 = 4$$

$$\text{Output}[1, 2] = 3(1) + 6(0) + 1(1) + 1(-1) = 3 + 0 + 1 - 1 = 3$$

Final Output Matrix:

$$\begin{bmatrix} -8 & 4 & 1 \\ 5 & 4 & 3 \end{bmatrix}$$